

3-Ethyl-3-methylpentane: A Technical Safety and Handling Guide for Researchers

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Compound of Interest

Compound Name: 3-Ethyl-3-methylpentane

CAS No.: 1067-08-9

Cat. No.: B092670

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An In-depth Guide for Researchers, Scientists, and Drug Development Professionals

This technical guide provides a comprehensive overview of the safety data, handling precautions, and toxicological profile of **3-Ethyl-3-methylpentane** (CAS No. 1067-08-9). The information is intended to support risk assessment and ensure safe laboratory practices for professionals working with this compound. While specific experimental data for **3-Ethyl-3-methylpentane** is not always available in publicly accessible literature, this guide outlines the standard methodologies used for hazard evaluation based on internationally recognized guidelines.

Physicochemical and Hazard Identification

3-Ethyl-3-methylpentane is a flammable, colorless liquid.^[1] It is a branched alkane and an isomer of octane. Due to its non-polar nature, it is insoluble in water but soluble in organic solvents.

Table 1: Physicochemical Data for 3-Ethyl-3-methylpentane

Property	Value	Source
Molecular Formula	C8H18	[2]
Molecular Weight	114.23 g/mol	[2]
CAS Number	1067-08-9	[2]
Boiling Point	118 °C	ECHEMI
Flash Point	6 °C (closed cup)	ECHEMI
Density	0.711 g/cm ³	Chemsrsrc
Vapor Pressure	20.2 mmHg at 25°C	Chemsrsrc
Water Solubility	Insoluble	

Table 2: GHS Hazard Classification

Hazard Class	Category	Hazard Statement
Flammable Liquids	2	H225: Highly flammable liquid and vapor[1]
Skin Corrosion/Irritation	2	H315: Causes skin irritation[1]
Aspiration Hazard	1	H304: May be fatal if swallowed and enters airways
Specific Target Organ Toxicity (Single Exposure)	3	H336: May cause drowsiness or dizziness
Hazardous to the Aquatic Environment, Acute Hazard	1	H400: Very toxic to aquatic life
Hazardous to the Aquatic Environment, Chronic Hazard	1	H410: Very toxic to aquatic life with long lasting effects

Experimental Protocols for Safety Evaluation

Detailed experimental reports for **3-Ethyl-3-methylpentane** are not readily available in the public domain. However, the safety data reported in Safety Data Sheets (SDS) are typically

derived from studies conducted according to standardized guidelines, such as those established by the Organisation for Economic Co-operation and Development (OECD). The following sections describe the likely methodologies used to assess the key hazards of this substance.

Acute Oral Toxicity

The acute oral toxicity of a substance is typically determined using one of the OECD test guidelines: TG 420 (Fixed Dose Procedure), TG 423 (Acute Toxic Class Method), or TG 425 (Up-and-Down Procedure).[3] The Acute Toxic Class Method (OECD TG 423) is a commonly used approach.[3][4]

Methodology (Based on OECD TG 423):

- **Test Animals:** The test is typically conducted using a small number of rats (usually females), with 3 animals per step.[3]
- **Dosing:** The substance is administered as a single oral dose via gavage.[5] The starting dose is selected from a series of fixed dose levels (5, 50, 300, 2000, or 5000 mg/kg body weight).[3]
- **Observation:** Animals are observed for mortality and clinical signs of toxicity for up to 14 days.[5]
- **Endpoint:** The method allows for the classification of the substance into a GHS toxicity category based on the observed mortality at different dose levels.[5] A definitive LD50 value is not the primary endpoint.

Skin Irritation

In vitro methods are now the standard for determining skin irritation potential, avoiding the use of animal testing. The OECD TG 439 (In Vitro Skin Irritation: Reconstructed Human Epidermis Test Method) is the relevant guideline.[6][7]

Methodology (Based on OECD TG 439):

- **Test System:** A three-dimensional reconstructed human epidermis (RhE) model is used, which mimics the properties of human skin.[8]

- Application: The test substance is applied topically to the surface of the RhE tissue.[8]
- Incubation: Following a defined exposure period, the tissue is rinsed and incubated.
- Viability Assessment: Cell viability is determined using a colorimetric assay, such as the MTT assay.[8]
- Endpoint: A substance is classified as a skin irritant (GHS Category 2) if the mean tissue viability is reduced to $\leq 50\%$ of the negative control.[6][8]

Flammability

The flammability characteristics of a liquid, such as its flash point and flammability limits, are determined using standardized methods like those from ASTM International.

Methodology (Based on ASTM E681 for Flammability Limits):

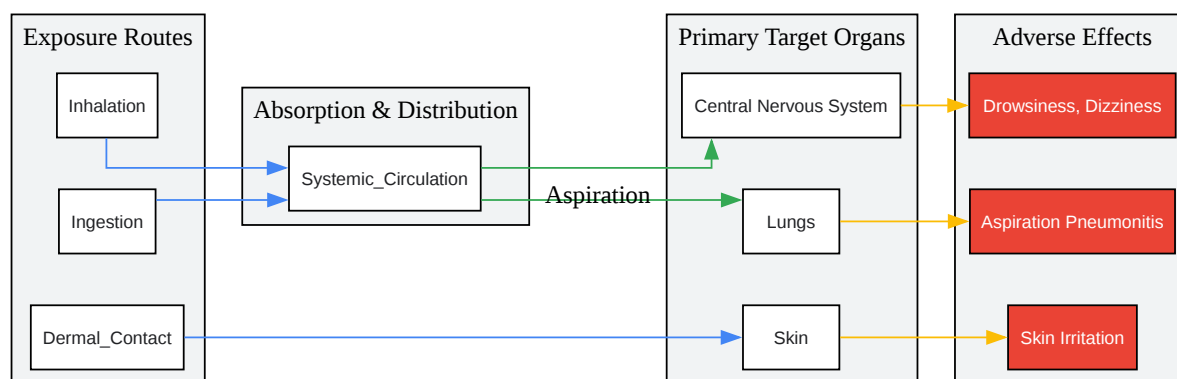
- Apparatus: A closed spherical glass flask (typically 5-L) is used, equipped with a pressure sensor and an ignition source.[9]
- Procedure: A known concentration of the substance's vapor is introduced into the flask with air.
- Ignition: An attempt is made to ignite the mixture with a spark.
- Endpoint: The Lower Flammable Limit (LFL) and Upper Flammable Limit (UFL) are the lowest and highest concentrations of the vapor in air that will propagate a flame.[10] The criterion for ignition is typically a specified pressure rise within the vessel.[9]

Toxicological Profile and Handling Precautions

Mechanism of Toxicity

While a specific signaling pathway for **3-Ethyl-3-methylpentane** has not been elucidated, the toxicity of volatile hydrocarbons is generally attributed to their physicochemical properties. As lipophilic compounds, they can partition into the lipid bilayers of cell membranes, disrupting their structure and function. This can lead to increased membrane permeability and impairment of membrane-bound proteins.

The primary toxicological effects of petroleum distillates like **3-Ethyl-3-methylpentane** are depression of the central nervous system (CNS) and pulmonary damage.[2] CNS effects are likely due to the disruption of neuronal membranes. Pulmonary damage is a major concern following aspiration, where the substance can interfere with surfactant function in the lungs.[2]



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General mechanism of hydrocarbon toxicity.

Handling and Storage

Engineering Controls:

- Handle in a well-ventilated area, preferably in a chemical fume hood.
- Use explosion-proof electrical, ventilating, and lighting equipment.[1]
- Ground and bond containers and receiving equipment to prevent static discharge.[1]

Personal Protective Equipment (PPE):

- Eye/Face Protection: Wear tightly fitting safety goggles.
- Skin Protection: Wear protective gloves (e.g., nitrile rubber) and a lab coat. Flame-retardant clothing is recommended.

- Respiratory Protection: If ventilation is inadequate, use a suitable respirator with an organic vapor cartridge.

Storage:

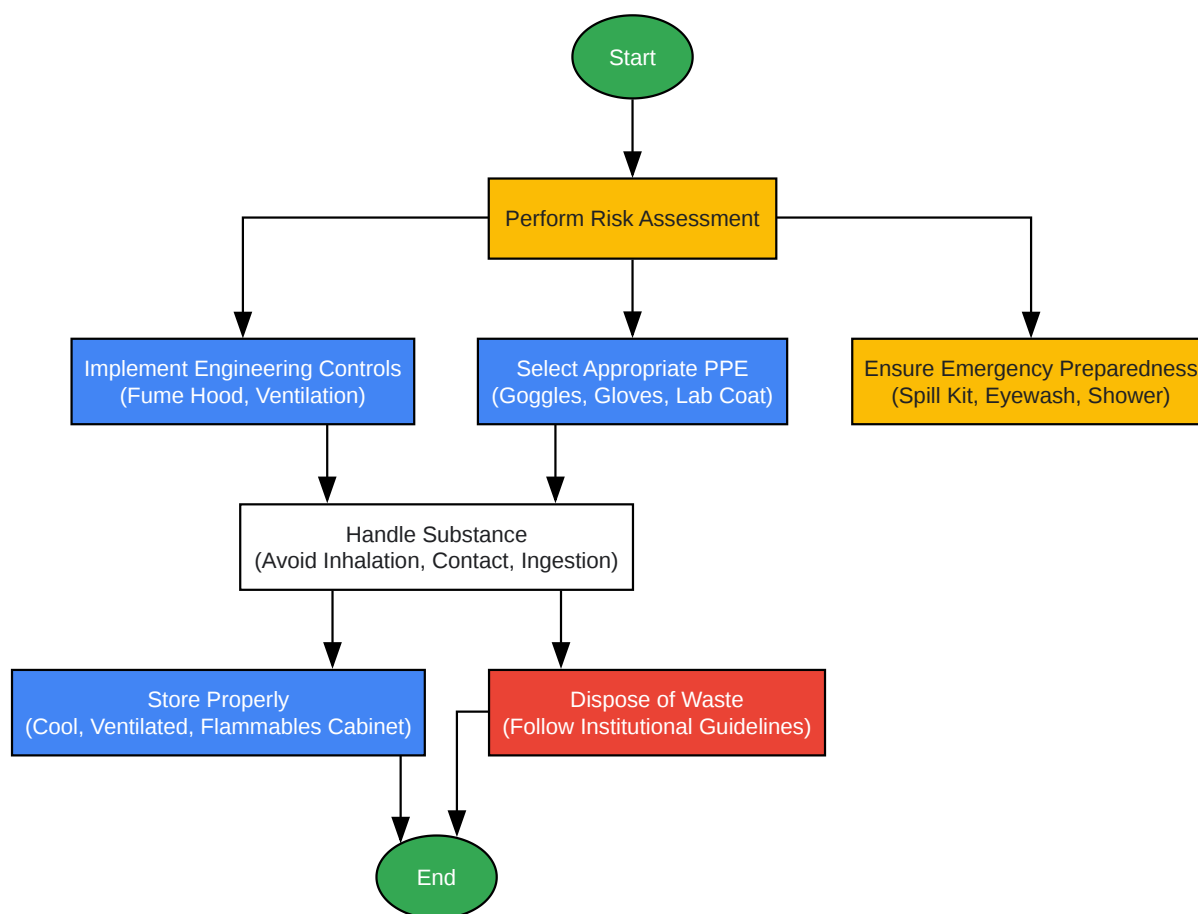
- Store in a tightly closed container in a dry, cool, and well-ventilated place.^[1]
- Keep away from heat, sparks, open flames, and other ignition sources.^[1]
- Store locked up.

First Aid Measures

- Inhalation: Move the person to fresh air. If not breathing, give artificial respiration. Consult a physician.
- Skin Contact: Immediately wash off with soap and plenty of water. Consult a physician.
- Eye Contact: Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.
- Ingestion: Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Rinse mouth with water. Consult a physician immediately. Aspiration hazard is a major concern.

Safety Workflow

The following diagram outlines a logical workflow for the safe handling of **3-Ethyl-3-methylpentane** in a laboratory setting.



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Workflow for safe handling of **3-Ethyl-3-methylpentane**.

Conclusion

3-Ethyl-3-methylpentane is a hazardous chemical that requires careful handling due to its high flammability, skin irritancy, and aspiration toxicity. While specific, publicly available experimental data on this compound is limited, adherence to the general safety principles for volatile hydrocarbons and the use of appropriate engineering controls and personal protective equipment are essential for minimizing risk. This guide provides a framework for understanding the potential hazards and implementing safe laboratory practices. Researchers should always consult the most recent and complete Safety Data Sheet provided by the supplier before working with this or any other chemical.

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